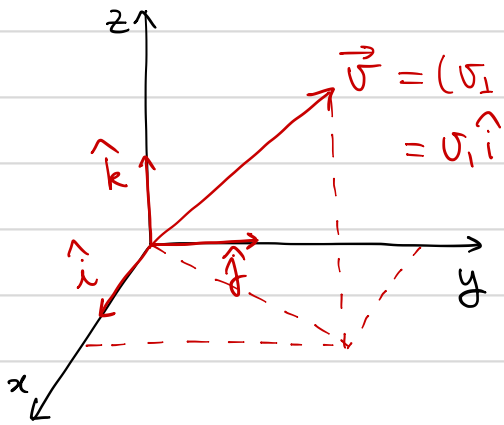
Review of vectors (12.3-4)



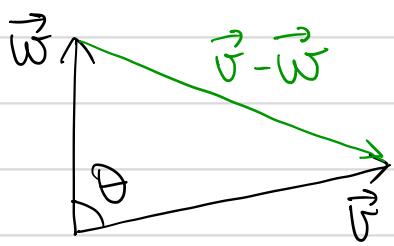
$$\vec{v} = (v_1, v_2, v_3) = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$$

$$\hat{i} = (1,0,0)$$
$$\hat{j} = (0,1,0)$$
$$\hat{k} = (0,0,1)$$

$$\text{length}^2 = |\vec{v}|^2 = \sqrt{v_1^2 + v_2^2}^{\,2} + v_3^2 = v_1^2 + v_2^2 + v_3^2 = \vec{v}\cdot\vec{v}$$

Dot product:

$$\vec{v}\cdot\vec{w} = v_1w_1 + v_2w_2 + v_3w_3 = |\vec{v}|\cdot|\vec{w}|\cdot\cos\theta$$



Why:

Law of cosines (see supplementary notes)

$$|\vec{v}-\vec{w}|^2 \overset{!}{=} |\vec{v}|^2 + |\vec{w}|^2 - 2|\vec{v}||\vec{w}|\cos\theta$$

$$\parallel$$

$$(\vec{v}-\vec{w})\cdot(\vec{v}-\vec{w}) = \vec{v}\cdot\vec{v} + \vec{w}\cdot\vec{w} - 2\vec{v}\cdot\vec{w}$$

Cross product:

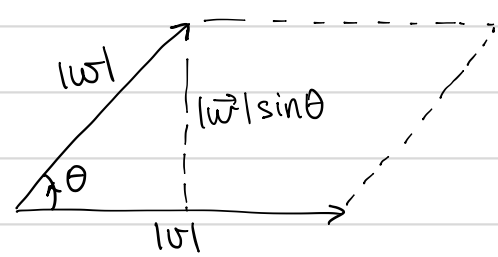
$$\vec{v}\times\vec{w} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} \begin{matrix} \hat{i} & \hat{j} \\ v_1 & v_2 \\ w_1 & w_2 \end{matrix}$$

$$= (v_2w_3 - v_3w_2)\hat{i} + (v_3w_1 - v_1w_3)\hat{j} + (v_1w_2 - v_2w_1)\hat{k}$$

Facts:

① $|\vec{v}\times\vec{w}| = |\vec{v}||\vec{w}|\sin\theta$

= Area of paralelogram

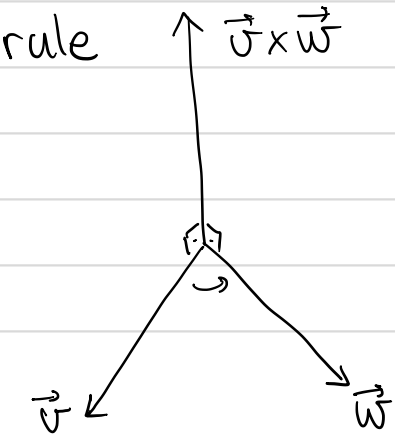


Why:

boring computation

$$|\vec{v}\times\vec{w}|^2 = |\vec{v}|^2|\vec{w}|^2 - (\vec{v}\cdot\vec{w})^2 = |\vec{v}|^2|\vec{w}|^2(1-\cos^2\theta) = |\vec{v}|^2|\vec{w}|^2\sin^2\theta \quad //$$

② Right hand rule

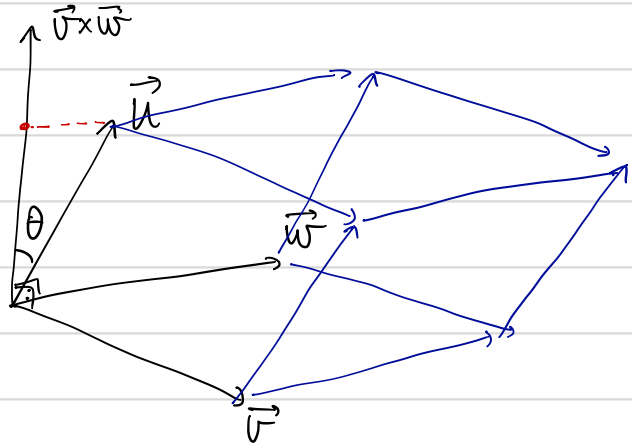


$$\vec{v}\cdot(\vec{v}\times\vec{w}) = 0$$
$$\vec{w}\cdot(\vec{v}\times\vec{w}) = 0$$

easy calculation

Determinant = ± Volume of box.

3D



$$\det\begin{bmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{bmatrix} =$$

$$= u_1(v_2w_3 - v_3w_2) + u_2(v_3w_1 - v_1w_3) + u_3(v_1w_2 - v_2w_1)$$

$$= \vec{u}\cdot(\vec{v}\times\vec{w})$$

$$= \underbrace{|\vec{u}|\cdot\cos\theta}_{\pm\text{height}} \cdot \underbrace{|v\times\vec{w}|}_{\text{area parallelogram}}$$

$$= \pm(\text{Volume of box})$$

Sign is minus if $\vec{u}$ and $\vec{v}\times\vec{w}$ are in opposite sides of the plane spanned by $\vec{v}$ and $\vec{w}$.

2D

$$\text{Area} = \pm\det\begin{bmatrix} u_1 & u_2 \\ v_1 & v_2 \end{bmatrix}$$

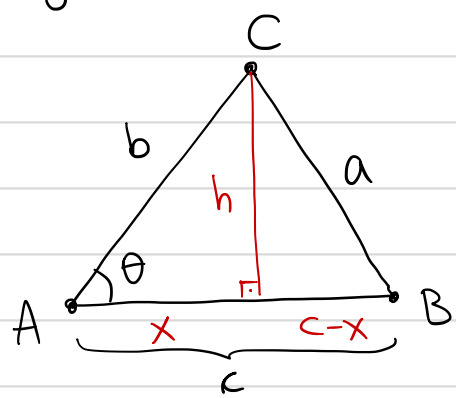
Why: $\vec{u},\vec{v}$ in the xy-plane

$$\Rightarrow \vec{u}\times\vec{v} = \det\begin{pmatrix} u_1 & u_2 \\ v_1 & v_2 \end{pmatrix}\hat{k}$$

$\rightarrow$

Supplementary Notes

Law of Cosines
(= Pythagorean theorem for angles other than 90°)



$$\boxed{a^2 = b^2 + c^2 - 2bc\cos\theta}$$

Proof:

$$\begin{cases} x^2 + h^2 = b^2 & (1) \\ (c-x)^2 + h^2 = a^2 & (2) \\ b\sin\theta = h & (3) \end{cases}$$

$$c^2 - 2cx + x^2 + h^2 \overset{(2)}{=} a^2$$

$$\overset{(1)}{=}$$

$$c^2 - 2c\sqrt{b^2 - h^2} + (b^2 - h^2) + h^2$$

$$\overset{(3)}{=}$$

$$c^2 - 2c\sqrt{b^2(1 - \sin^2\theta)} + b^2$$

$$=$$

$$c^2 - 2cb\cos\theta + b^2 \qquad \blacksquare$$